



Somesh Rahul <somesh.rahul33@gmail.com>

Lehigh Presentation

1 message

Rhodes Baker <rhodesbaker@gmail.com>
To: somesh.rahul33@gmail.com

Mon, Oct 13, 2025 at 11:04 AM

Somesh Rahul and I were invited to present at Lehigh University on two occasions — first on "Ambient Computing" and later on 'Designing for Zero UI.' The first session was well-received by students, professors, and visiting industry practitioners; based on that response, we were invited back the following year. Attendance and engagement were strong, and several attendees followed up with us afterward.

The professor commented that they have an imbalance between Industrial Applications and Theory, and could use more practical use presentations. The topics were not typically instructed on within the CS Department at the time. They were very excited to have us and between us and the Pizza the room was packed on both occasions.

The first session took place on 10/18/2016 and the slide deck is attached below.

Thank you,
Rhodes B. Baker

 **Ambient Computing and Zero UI - 10.18.2016.pdf**
2728K

Subject: Invitation to Speak at Lehigh University

Date: Monday, September 19, 2016

From: Office of Student Affairs instuaff@lehigh.edu

To: Somesh Rahul srahul@lutron.com; Rhodes Baker rbaker@lutron.com

Dear Mr. Rahul and Mr. Baker,

On behalf of the student body at Lehigh University, the Office of Student Affairs would like to invite you to our campus to deliver a lecture for our students.

We have been following the innovative work happening at Lutron Electronics with great interest. Your expertise in the area of **Ambient Computing** would be incredibly insightful for our students, particularly those in our engineering, computer science, and product design programs. Understanding the evolution of computing and designing for these emerging platforms is a highly relevant topic on campus.

We would be honored if you would consider presenting your talk on **Tuesday, October 18, 2016**. We are flexible, however, and would be happy to find an alternative date that suits your schedules.

Thank you for considering this invitation. We believe your perspective would be a valuable addition to our students' learning experience and look forward to hearing from you.

Sincerely,

The Office of Student Affairs
Lehigh University
(610) 758-3890



Ambient Computing

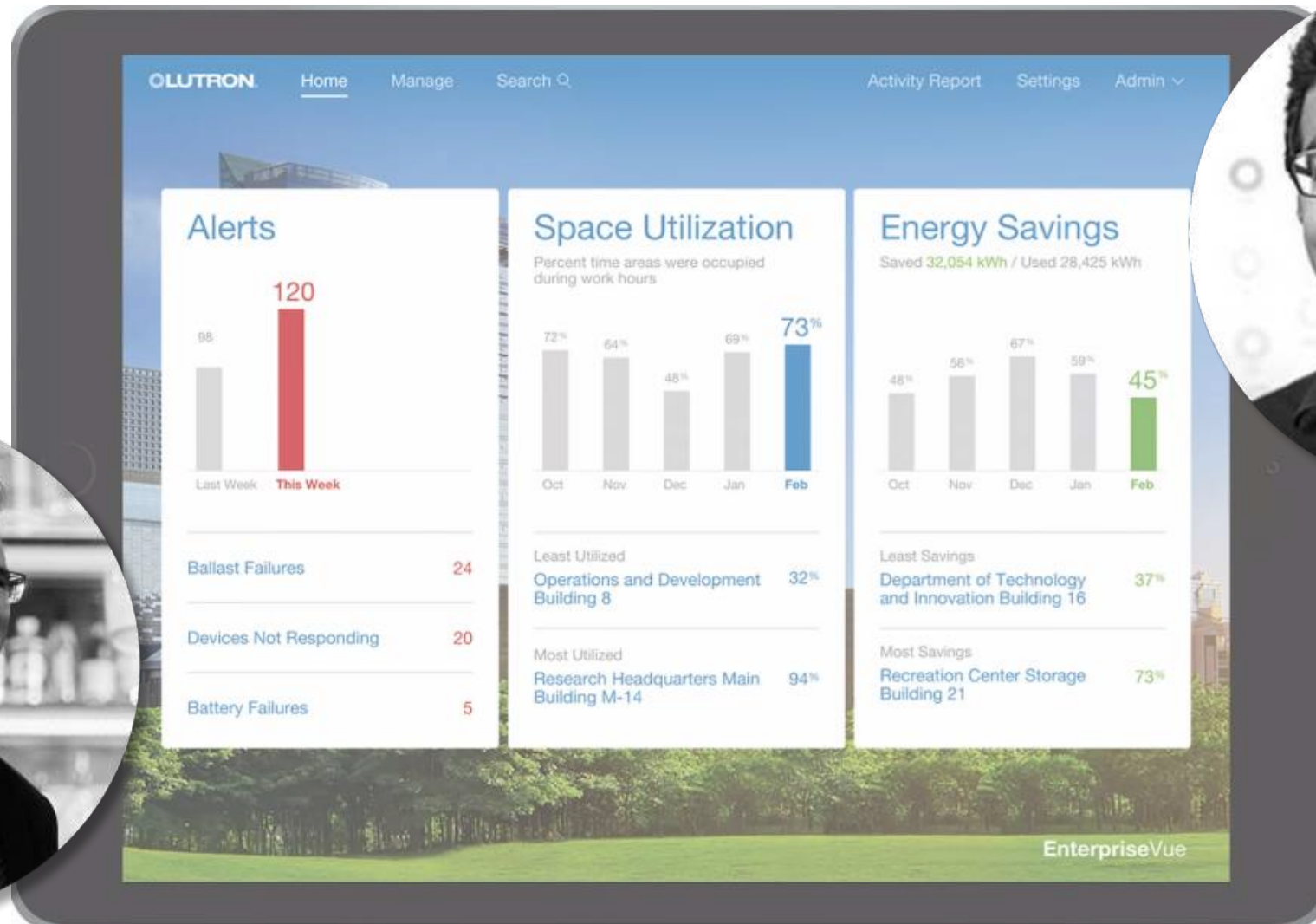
Rhodes Baker

Manager of Residential Technology

Somesh Rahul

Sr. UX Engineer

Introductions



Ambient Computing Basics and Evolution

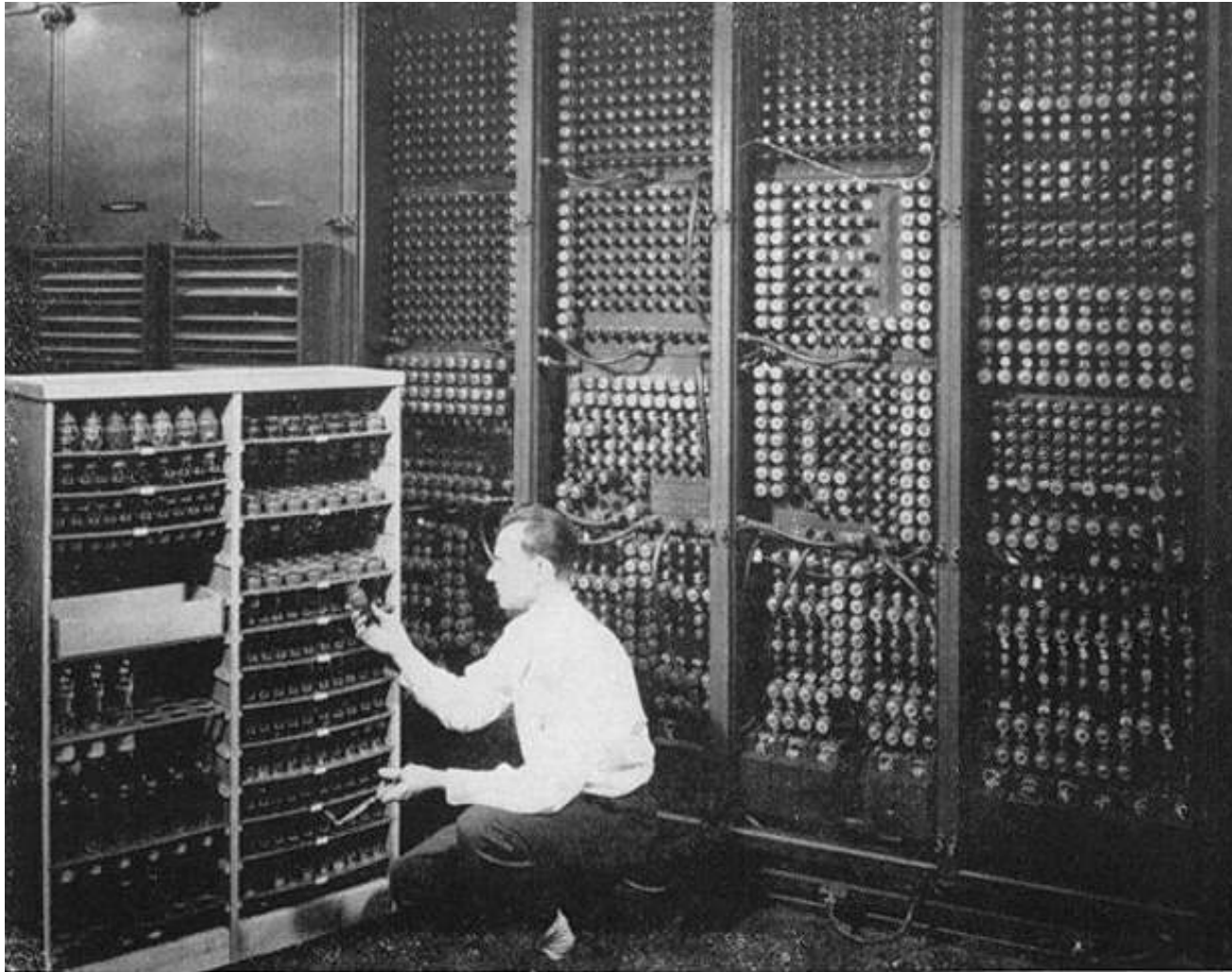
1. Evolution
2. What is Ambient Computing
3. Applications
4. Why Now
5. Designing for Ambient Computing

Benefit Friction Causes Adoption Delay

1. Any remote control, car dashboard, screen, or app requires some level of learning, or friction
2. The less friction before realizing the benefit the better the solution



History



Desktop



Laptop – Portable Computers



Smart Phones – Computer in Palms



Wearables – Computers on You



Voice – Computers around You



Alexa



Siri



Google Now



Cortana

Benefit Friction - Quantitative

Era	Task	
 90's – early 2000's	Turn on PC Open AOL Connect to dial up Open browser Go to web page Total:	6 min
 2000s	Wake up laptop Go to Firefox Go to web page Total:	1 min

Benefit Friction - Quantitative

Era	Task	Time
 2010s	Take phone out of pocket	
	Unlock with fingerprint	
	Open app	
	Total:	15 sec



2015

Look at wrist

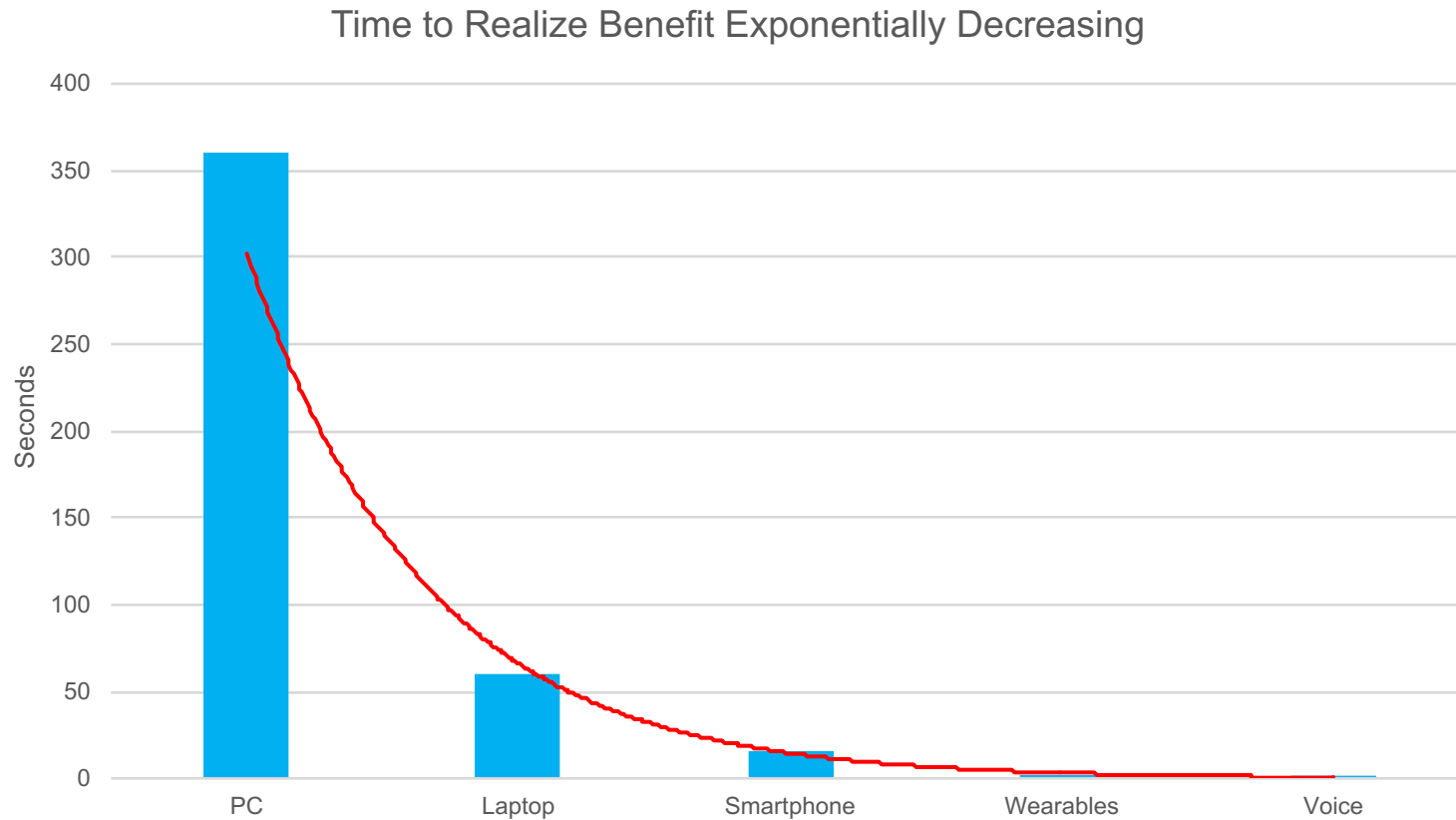
Total:

3 sec

Benefit Friction - Quantitative

Era	Task	Time
	Ask what you don't know	
2018s	Total:	1 sec

Friction Benefit Exponentially Decreasing

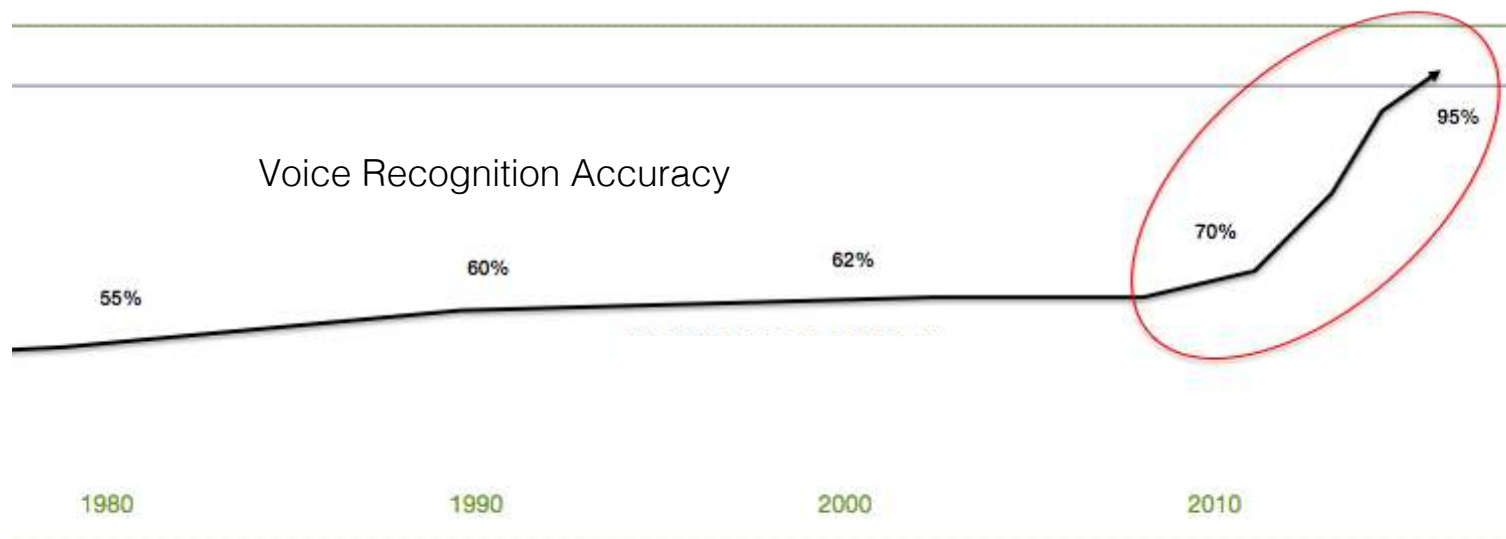


What is Ambient Computing?

1. Ambient intelligence/computing refers to electronic environments that are sensitive and responsive to the presence of people. – Concept developed by Eli Zelkha and his team at Palo Alto Ventures
2. Works on the principle of Zero UI - “natural gestures, body movements, words, maybe even thoughts can trigger the results we want from a device rather than having to interact in an abrupt way and become absorbed in a screen.” – Andy Goodman, coined the phrase

Why Now?

1. High-speed internet access
2. Cloud computing
3. Advancements in personal computing
4. User comfort
5. Battery enhancements



Sensors and Automation



Lutron Occupancy Sensors

Advanced Sensors

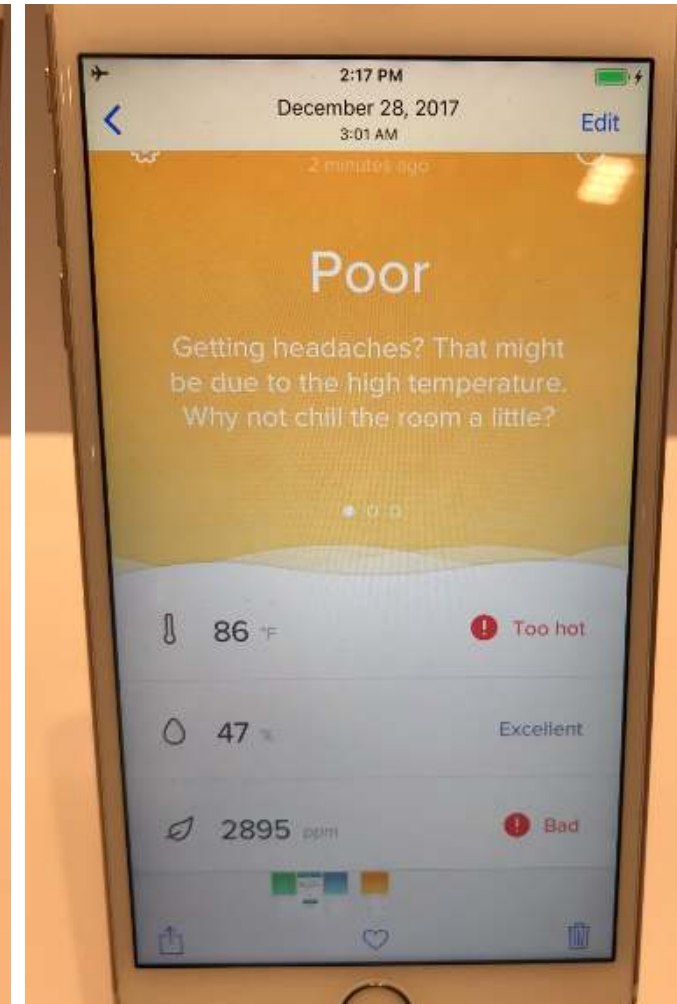
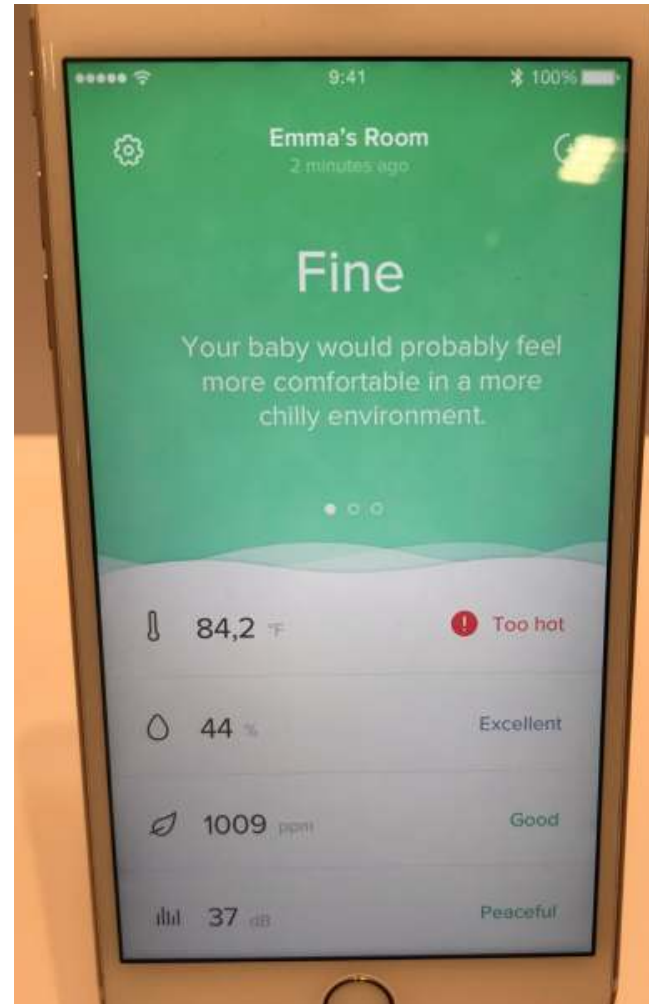
1. Multi or “Omni” sensors measure multiple physical states, report to extensible 3rd party platforms like SmartThings, IFTTT



Aeotec MultiSensor6

Air Quality Sensors

1. Detect Everything from Humidity to Pollutant PPM



Sensors Can be Simple - Loreial UV Sense

1. Changes color when you've had too much sun.

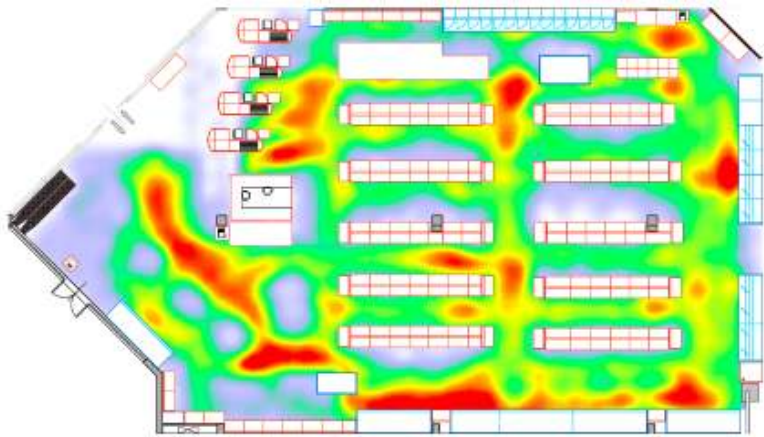


Sensors + AI



Indoor Positioning

1. Asset tracking of objects and individuals inside a building
2. Instantaneous (where is a specific object) and historic (how many objects over time)
3. Delivered through a combination of sensing technology on objects, or built into mobile smartphones



RetailerIN Analytics



Stanley Healthcare

Voice – Amazon Echo

1. “What’s the weather”
2. “Play Frank Sinatra”
3. “Turn the lights off”
4. “Dim the lights”
5. “Ask Dominos to order a pizza”
6. “Alexa, good morning”



Voice In All The Things



iDevices



GE Lighting

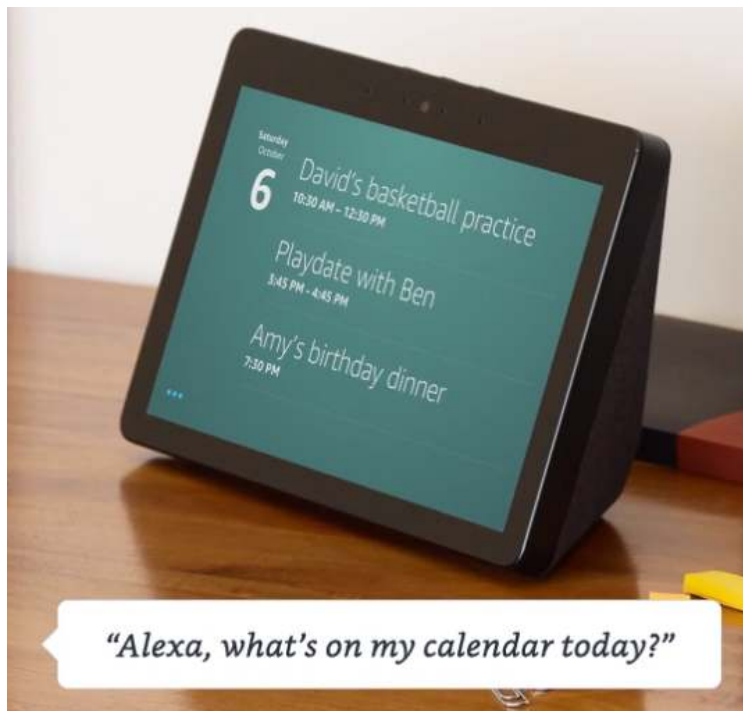


Johnson Controls

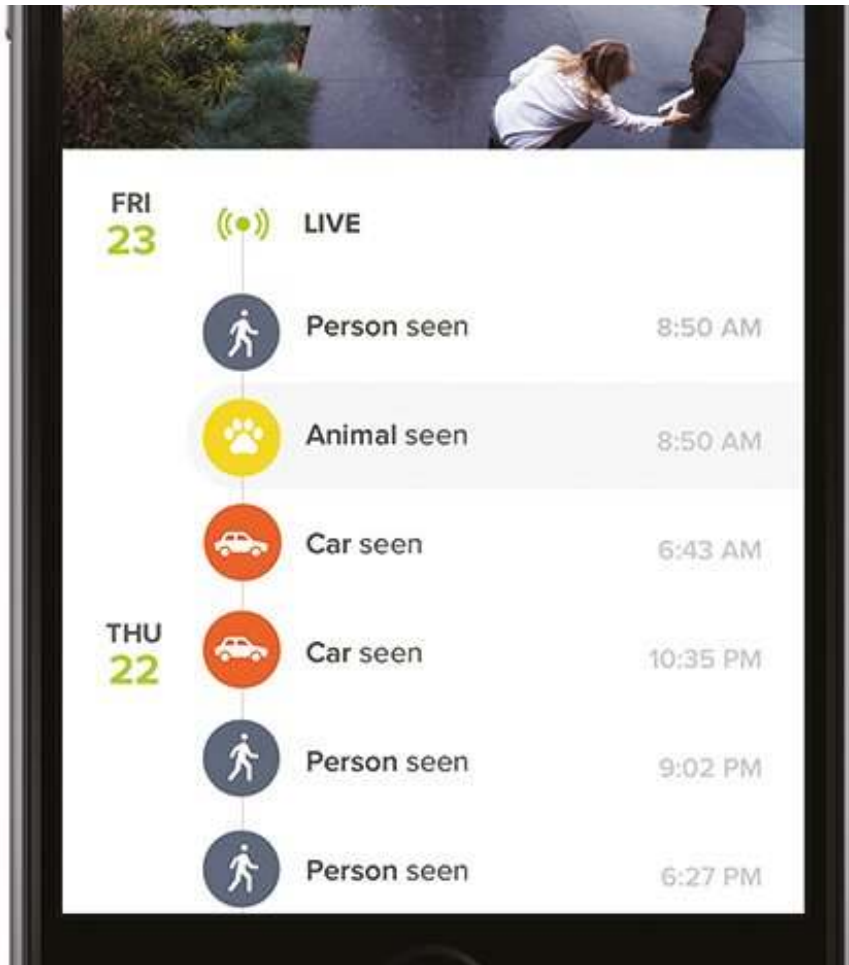


GE Lighting

Voice + Screen



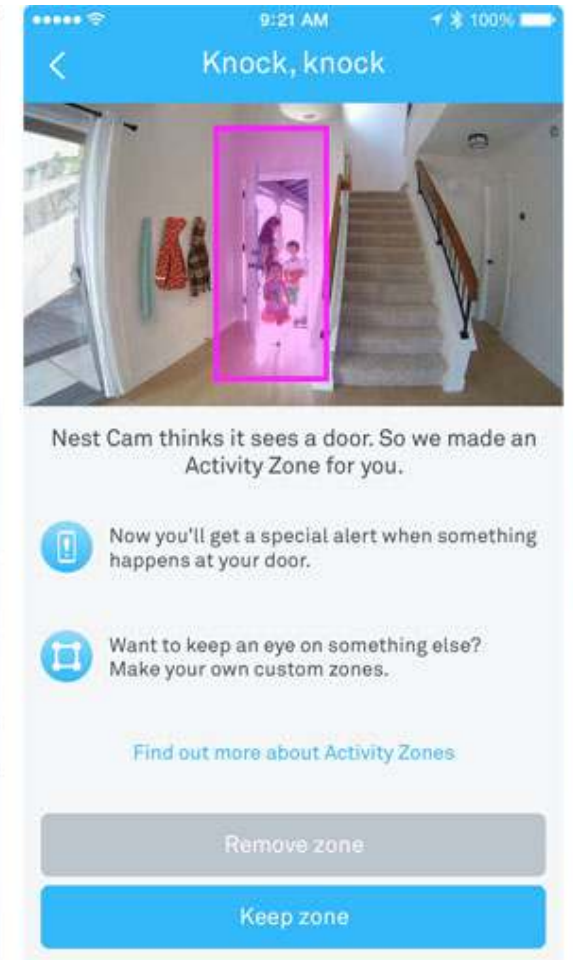
Video Analytics



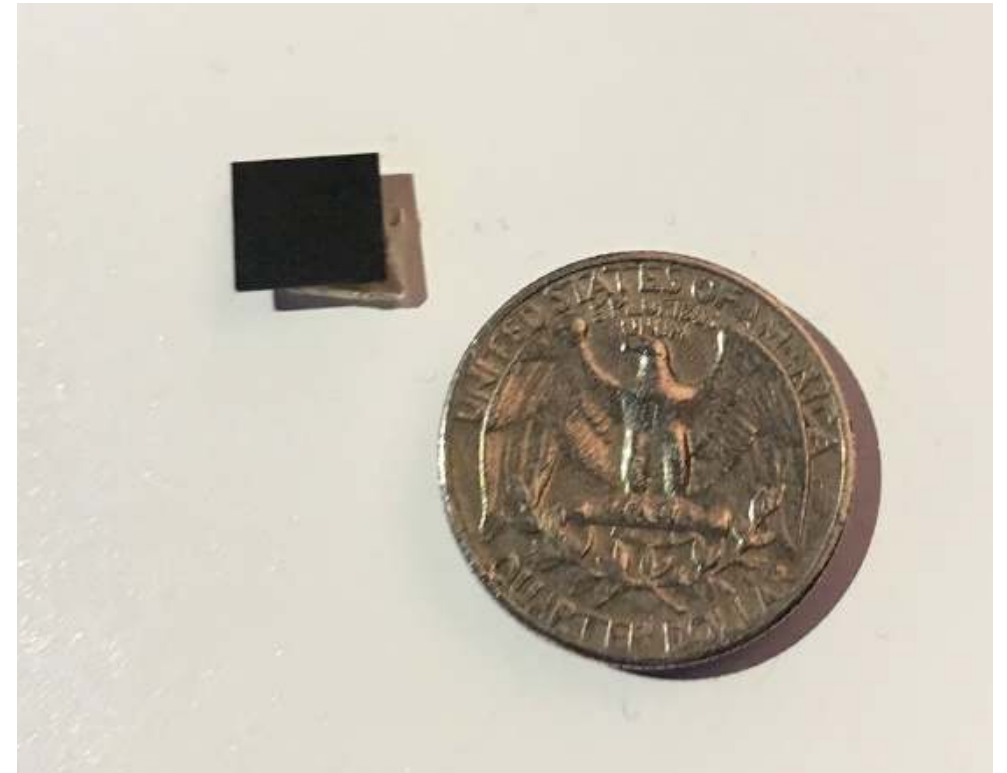
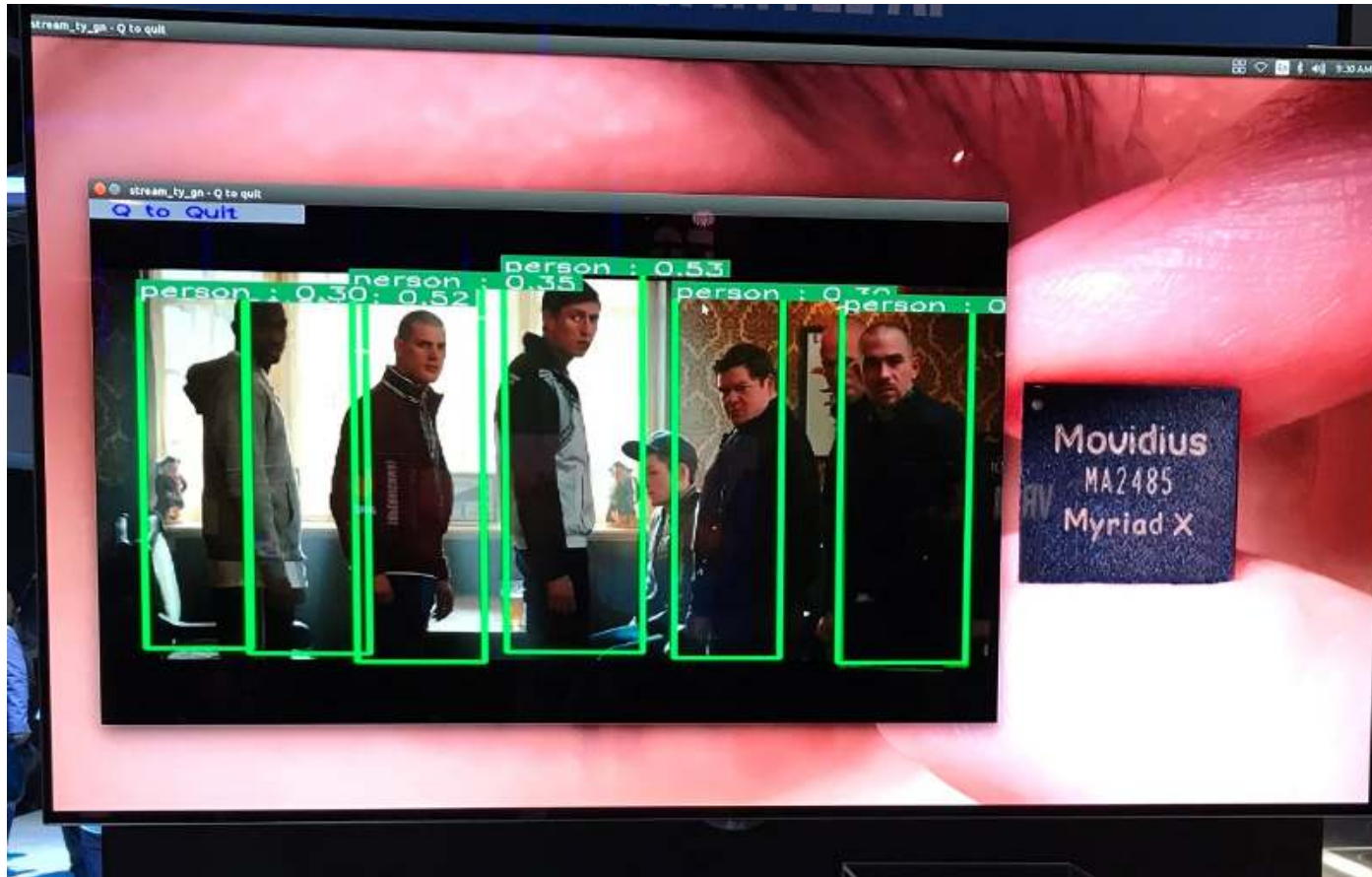
Netatmo



Nest



NXP Video Analytics



Video Processing – Casino Edition

1. Facial recognition.
2. Records number and suit on cards
3. Expression recognition
4. Augments security team, does not replace



Audio – Hearables



From the movie Her

Hearables

1. Music, microphone, calls
2. Rechargeable battery
3. Noise cancelling
4. Biometric sensors
5. Built-in Siri



Samsung Gear IconX



Apple AirPods

Physical Movements



Apple Pay on Apple Watch

Wearables Update – Apple Watch

Notifications



Lutron

Contextual



Starwood Hotels

At a Glance and Health Tracking



Designing For Ambient Computing

1. Think away from linear workflows, and towards multi-dimensional thought process
2. Leverage computing, low cost sensors, etc.
3. Learn, adapt, and anticipate human needs
4. Device ecosystem



- Andy Goodman, Alex Capecelatro

From Screens to Scenes and Context

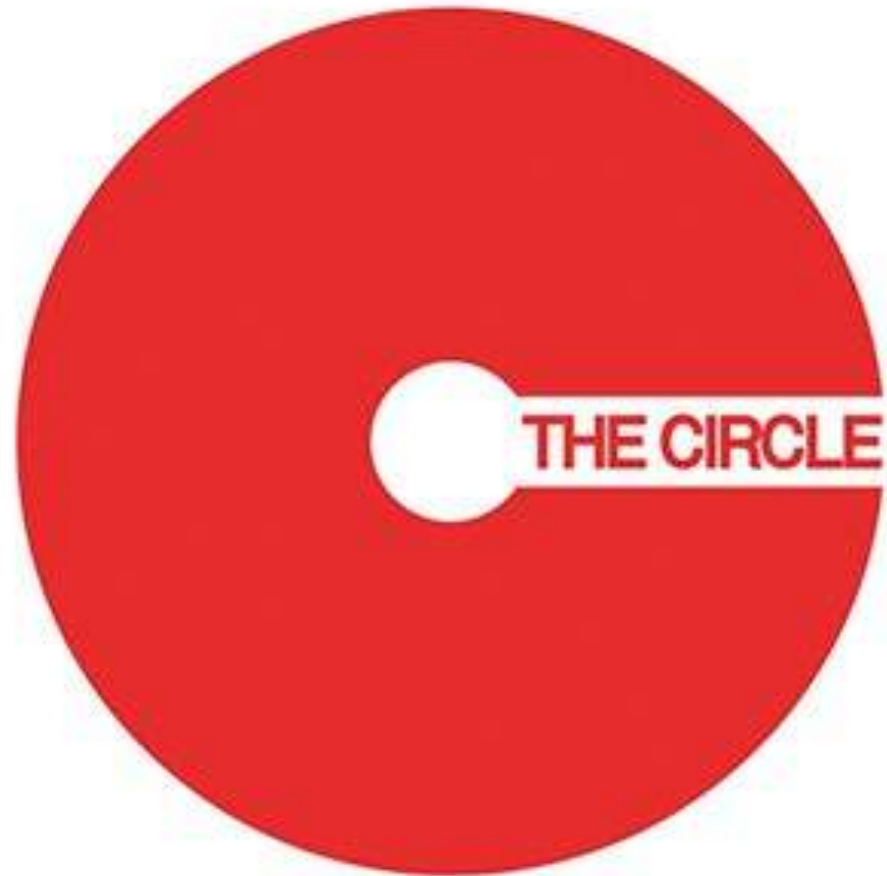


The Dark Side



The Dark Side

Knowing is good.
Knowing everything is better.



Future – Where Things are Headed

1. AI first approach
2. Device-less state
3. Pervasive



Instead of asking us to learn a new interface, it (the interface) works with our normal everyday habits.”

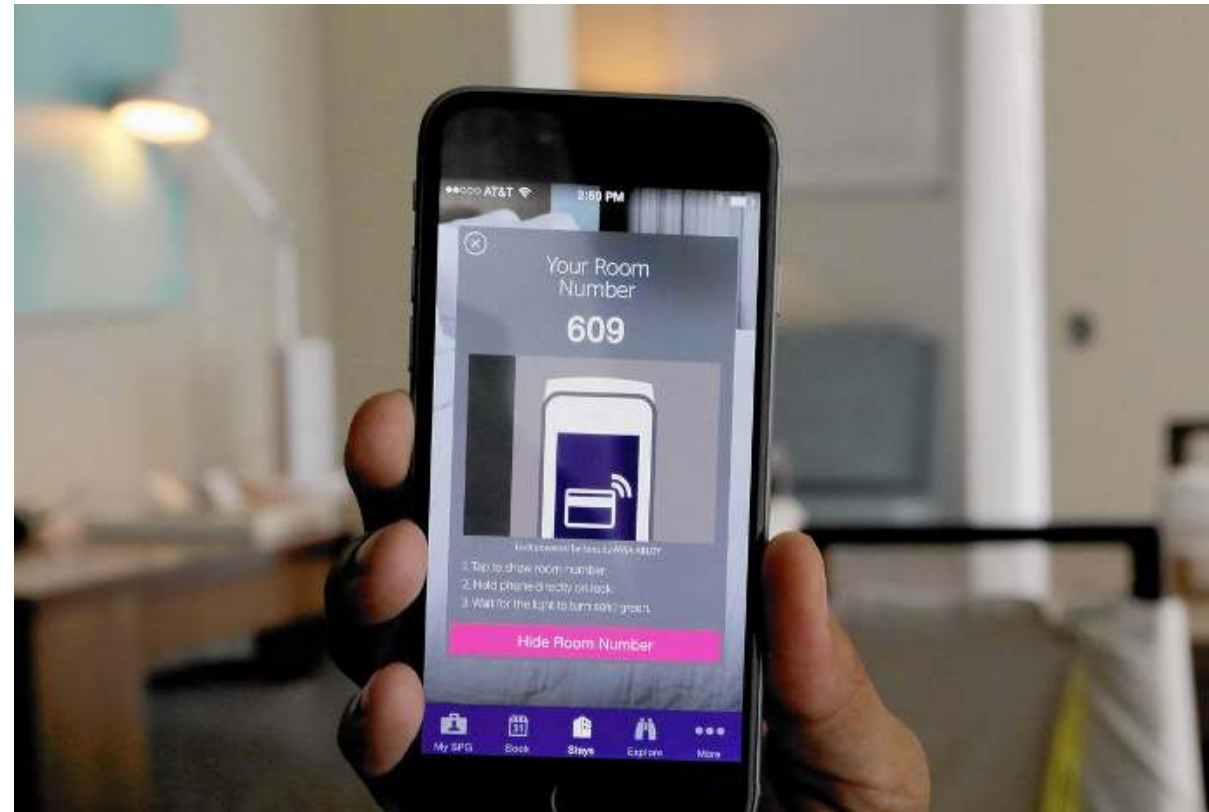
The Best Interface Is No Interface
by Golden Krishna

Thank You!

Amazon Dash Button



Hotel Check-In



Gesture Control – MOOV Ring Example



Fibaro Gestures

- Gestures can be used to control Lights, and Shades



Actionable Notifications

1. Contextual
2. Brief
3. Valuable
4. Single - touch



By Snapchat

1. Decreases friction of recording and  Spectacles



A New Dimension of Design

1. From what a user is trying to do right now in a linear, predictable workflow—designers need to think about what a user is trying to do right now in any possible workflow.- [John Brownlee](#)



Zero UI Design Exercise

Arriving home from work
Park in garage
Enter the house
Turn the lights on
Turn on the news
Change out of work clothes
Cook dinner
Eating dinner
Cleaning up
...

Example

Let's try to reimagine the BMW app now with Zero UI

Car Example

In the winter, getting in your car and driving to work.

Nest Thermostat

- Learns user's behavior
- Uses sensors and phone location to activate the away mode to save energy
- Encourages users to save energy via in-app and on-screen notifications



Changes from 10/18/18 Call – RB, SR

1. Wearables – what happened?
 - A. Apple made series 1 too complex – the app based UI took too long to navigate, load the apps, and download the data to populate the apps – this broke the promise of a zero friction experience.
 - B. Now... apple is much more focused on the clock face complications – quick glances at the face head, health tracking sensors operate automatically, and instant notifications.
2. Voice services into the timeline
 - A. Alexa everywhere
 - B. Voice as an entry to visual – Alexa with screen built in
3. Advancements...
 - A. Voice –
 - i. Whisper
 - B. Indoor positioning (commercial)
 - i. Tracking people to maximize conference room usage
 - C. Sensors
 - i. Ubiquity sensors
4. Convert to wide screen
5. Update font – “make it more interesting” – Somesh

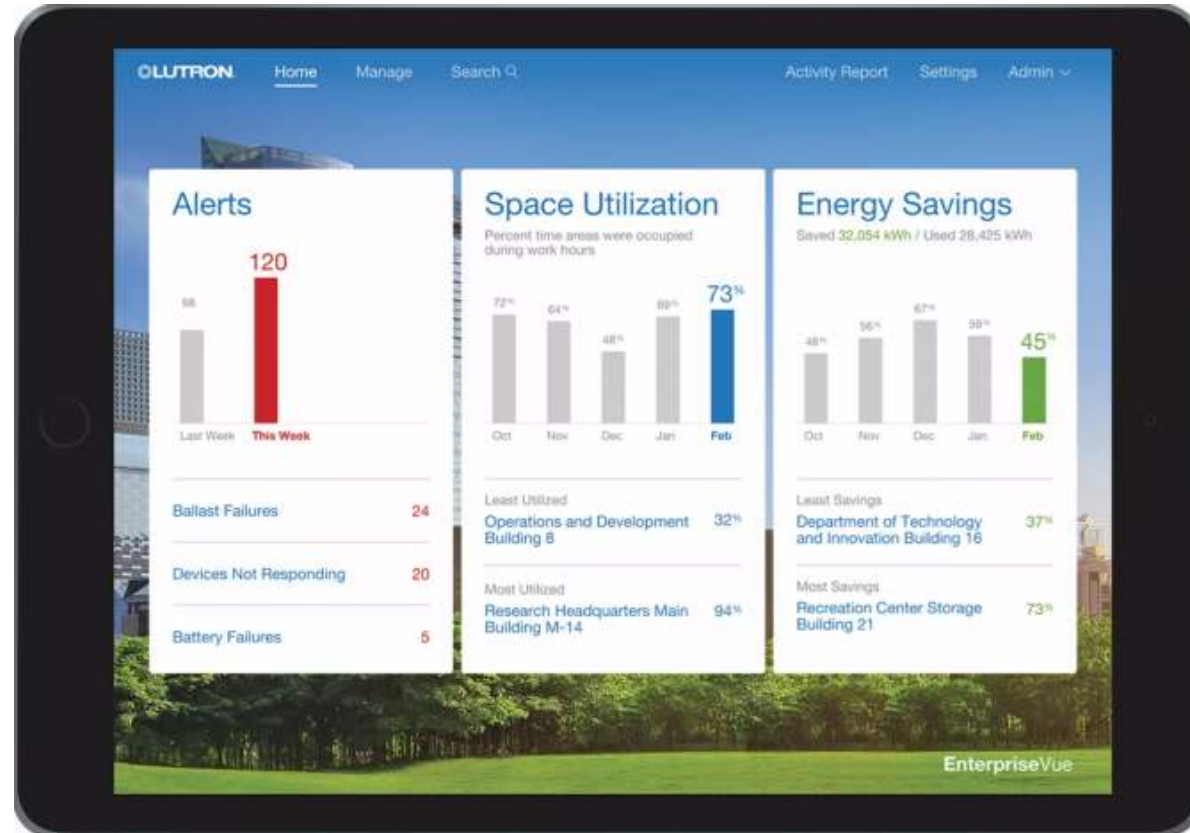
Introductions



Rhodes Baker



Somesh Rahul



Lutron – Enterprise Vue

History



1977

History



1996

History



2007, 2010

History



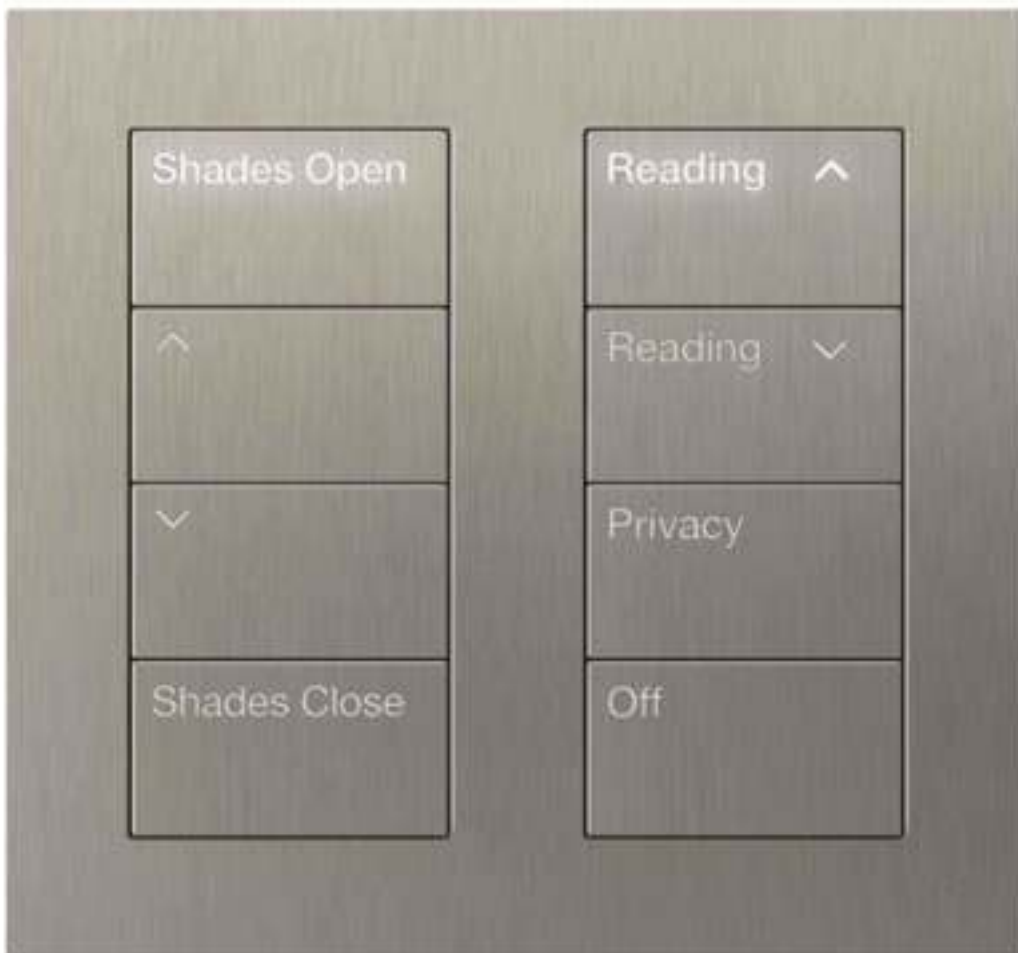
2015

IoT and Designing for Zero UI

Rhodes Baker
Somesh Rahul
Lutron Electronics

Evolution of User Interfaces

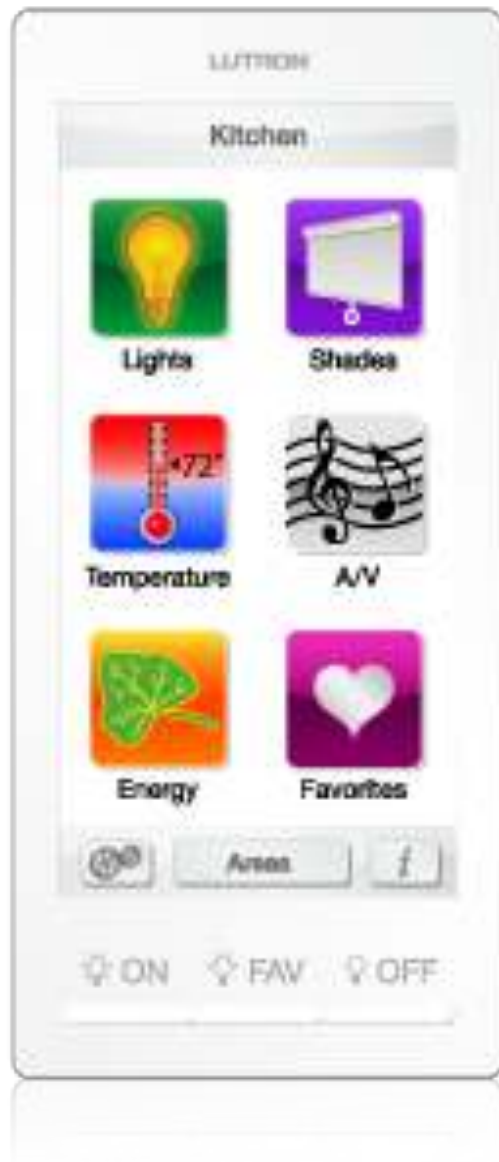
Physical Interfaces



http://www.gctel.com/images/television/mxv4ir_remote.jpg

http://i.i.cbsi.com/cnwk.1d/i/tim//2010/04/23/34069639_SS09.jpg

Digital Interfaces



http://www.info-utiles.fr/Images2016/Sevenhugs_SmartRemote_1.jpg

http://cdn1.autoexpress.co.uk/sites/autoexpressuk/files/styles/gallery_adv/public/tesla-model-s-3_0.jpg?itok=Cma2GwnR

Zero User Interfaces

- Definition, Dimension, etc.

Expanding on Zero UI

Why Now?

Accuracy

- alexa can recognize 90 out of 100 spoken words (english only)
- accurate sensors

Why Now?

Personal Computing

- phones (everyone owns them)

Why Now?

Computing Power

- Moore's Law

Different Zero UI Avenues

Voice

- alexa
- siri
- google home

AI

- Tesla auto pilot

Gesture Control

- Fibaro

Image Recognition

- nest

Virtual Reality

Augmented Reality

- holo lens
- tango

Embrace Zero UI

Principles

- Adapt to individual behaviors and habits
- Leverage data and computing
- Think in terms of activities instead of screens

Example

- Volkswagen Example

Example

- How will we do it with Zero UI

Instead of asking us to learn a new interface, it (the interface) works with our normal everyday habits.”

The Best Interface Is No Interface
by
Golden Krishna